5.5 Optimization

Name _____ Period ____ Date _____

AP Calculus 5.5 · Student sheet

I can…

- write the quantity to maximize or minimize as a function of one variable.
- find and justify the maximum or minimum.
- solve area, volume, number, and distance optimization problems.

Vocabulary · fill in the missing word

Objective function — the quantity to be _____ or minimized, written as a function

Constraint — an equation that relates the _____, used to write the objective in one variable

Part 1 · Fence it

40 ft of fence: a rectangle x ft wide and (20 − x) ft long.

Width x	2	5	8	10	12
Area					

Part 2 · Notes and examples (fill in each note, then work the example)

Step 1: draw a picture and name the variables; write the _____ function.

Step 2: use the _____ to write the objective function in one variable, and state its domain.

Example 1

Find the largest area of a rectangle with perimeter 40 ft.

Step 3: find the critical numbers, then justify with the endpoints or the _____ Derivative Test.

Example 2

A farmer has 400 ft of fence for a rectangle along a river (no fence on the river side). Find the largest area.

Example 3

Two positive numbers have a sum of 30. Find the largest product.

Example 4

An open box is made from a 12-in square by cutting squares of side x from the corners. Find the largest volume.

Example 5

Find the point on $y = \sqrt{x}$ closest to (3, 0).

Part 3 · Partner practice

Partner 1

Two positive numbers have a product of 36. Find the smallest sum.

Partner 2

(AP® Practice) A rectangle has a perimeter of 24. Its maximum possible area is

(A) 36 (B) 32 (C) 144 (D) 24